

claude-wsl-ubuntu / 4f364eb7-a600-4a2c-9b37-e5a313b6db9d

Metadata

```
- Source: `claude-wsl-ubuntu`
- Kind: `claude`
- Source file: `\\wsl.localhost\Ubuntu\home\avidullu\.claude\projects\home-avidullu-projects-khelsutra-guru\4f364eb7-a600-4a2c-9b37-e5a313b6db9d\subagents\workflows\wf_fa7a2077-3a9\agent-a8c625abef0cba8dc.jsonl`
- SHA-256: `5df73fff18db5154abdd64a39bd58bf87df40791e223236c28afe2493fd96148`
- Source modified: `2026-06-29T10:51:50+00:00`
- Imported at: `2026-07-05T16:48:30+00:00`
- project: `wf_fa7a2077-3a9`
- session_id: `4f364eb7-a600-4a2c-9b37-e5a313b6db9d`
```

Transcript

1. user (2026-06-29T10:50:03.915Z)

Read these in /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer if present: docs/DOC_STATUS.md (the honest update?archive protocol), docs/RALLY_DETECTION_QUALITY_REPORT.md (the quality table + ship target), docs/NEXT_STEPS.md (open ship-blocking items). Report the doc-update protocol steps, the current quality numbers / ship target, and any open launch items. Note which files exist vs missing. Return data, not prose.

2. assistant (2026-06-29T10:50:07.845Z)

I'll read the requested documentation files. Let me first check if the directory and files exist.

3. user (2026-06-29T10:50:09.743Z)

```
total 1020
drwxr-xr-x  6 avidullu avidullu  4096 Jun 29 10:44 .
drwxr-xr-x 22 avidullu avidullu  4096 Jun 27 17:27 ..
-rw-r--r--  1 avidullu avidullu  3850 Jun 25 11:46 ABLATION_LAYER.md
-rw-r--r--  1 avidullu avidullu 52198 Jun 25 11:46 ANALYZERS_AND_RECONCILERS.md
-rw-r--r--  1 avidullu avidullu 19932 Jun 25 11:46 BACKLOG.md
-rw-r--r--  1 avidullu avidullu  9401 Jun 25 11:46 CLOUD_GPU_DRYRUN_GUIDE.md
-rw-r--r--  1 avidullu avidullu  1032 Jun 25 11:46 CODE_AUDIT_AND_TEST_HARDENING.md
-rw-r--r--  1 avidullu avidullu 29686 Jun 27 17:27 CODE_CLEANUP_AUDIT_2026-06-24.md
-rw-r--r--  1 avidullu avidullu  88508 Jun 25 11:46 CODE_MAP.md
-rw-r--r--  1 avidullu avidullu  17431 Jun 25 11:46 COMMERCIALIZATION.md
drwxr-xr-x  3 avidullu avidullu  4096 Jun 25 11:46 COMPUTE_DECOUPLED_SERVING
-rw-r--r--  1 avidullu avidullu  7053 Jun 25 11:46 CONVERT_GCS_VIDEOS_TO_HIGHLIGHTS.md
-rw-r--r--  1 avidullu avidullu 12328 Jun 25 11:46 CREATING_REELS.md
-rw-r--r--  1 avidullu avidullu 17142 Jun 25 11:46 DATA_IN_GCS.md
-rw-r--r--  1 avidullu avidullu 13204 Jun 25 17:57 DECISION_SNAPSHOT_REGRESSION.md
-rw-r--r--  1 avidullu avidullu   999 Jun 25 11:46 DEFERRED_HARDENING_PLAN.md
-rw-r--r--  1 avidullu avidullu 22433 Jun 25 11:46 DESKTOP_APP_PLAN.md
-rw-r--r--  1 avidullu avidullu 11972 Jun 25 11:46 DOC_STATUS.md
-rw-r--r--  1 avidullu avidullu  6895 Jun 25 11:46 EVALUATION.md
-rw-r--r--  1 avidullu avidullu  8929 Jun 25 11:46 EXECUTION_POLICY.md
-rw-r--r--  1 avidullu avidullu 15750 Jun 25 11:46 EXPERIMENT_HARNESS.md
-rw-r--r--  1 avidullu avidullu 35880 Jun 25 11:46 GOLDEN_REGRESSION_FIXTURES.md
-rw-r--r--  1 avidullu avidullu  1281 Jun 25 11:46 HARDENING_LOOP_HANDOFF.md
-rw-r--r--  1 avidullu avidullu  8334 Jun 25 11:46 HOSTING_PLAN.md
-rw-r--r--  1 avidullu avidullu  4259 Jun 25 11:46 HOW_RALLY_DETECTION_WORKS.md
-rw-r--r--  1 avidullu avidullu 14902 Jun 25 11:46 I18N_PLAN.md
-rw-r--r--  1 avidullu avidullu 10456 Jun 25 16:06 KHELSUTRA_PRODUCT_OVERVIEW.md
-rw-r--r--  1 avidullu avidullu 20278 Jun 25 11:46 MULTI_SIGNAL_FUSION_PLAN.md
-rw-r--r--  1 avidullu avidullu 22619 Jun 29 10:44 NEXT_STEPS.md
-rw-r--r--  1 avidullu avidullu 18727 Jun 29 10:44 NIGHTLY_ENGINE_REGRESSION.md
```

```

-rw-r--r-- 1 avidullu avidullu 15091 Jun 25 11:46 OWNED_MODEL_IMPLEMENTATION_PLAN.md
-rw-r--r-- 1 avidullu avidullu 34117 Jun 25 11:46 PACKAGING_PLAN.md
-rw-r--r-- 1 avidullu avidullu 11612 Jun 25 11:46 PERSONALIZATION_PLAN.md
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-rw-r--r-- 1 avidullu avidullu 48667 Jun 25 11:46 PLATFORM_ARCHITECTURE.md
-rw-r--r-- 1 avidullu avidullu 24322 Jun 25 11:46
POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md
-rw-r--r-- 1 avidullu avidullu 3739 Jun 25 11:46 PROJECT_DOC_TEMPLATE.md
-rw-r--r-- 1 avidullu avidullu 16364 Jun 25 11:46 PROMINENT_COURT_DETECTION.md
-rw-r--r-- 1 avidullu avidullu 61174 Jun 25 11:46 QUALITY_ITERATIONS.md
-rw-r--r-- 1 avidullu avidullu 6368 Jun 25 11:46 R1_MILESTONE_LAUNCH_REPORT.md
-rw-r--r-- 1 avidullu avidullu 14872 Jun 25 11:46 R2_EVAL_METRIC_DESIGN.md
-rw-r--r-- 1 avidullu avidullu 6392 Jun 29 10:44 RALLY_DETECTION_CODE_REVIEW.md
-rw-r--r-- 1 avidullu avidullu 8032 Jun 29 10:44 RALLY_DETECTION_FIXES_PLAN.md
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-rw-r--r-- 1 avidullu avidullu 16807 Jun 25 17:57 README.md
-rw-r--r-- 1 avidullu avidullu 16130 Jun 25 11:46 REAL_FOOTAGE_VALIDATION_REPORT.md
-rw-r--r-- 1 avidullu avidullu 8133 Jun 25 11:46 ROADMAP.md
-rw-r--r-- 1 avidullu avidullu 15808 Jun 25 11:46 SETUP_NEW_MACHINE.md
-rw-r--r-- 1 avidullu avidullu 11746 Jun 25 11:46 STORAGE_SHARING_MODEL.md
-rw-r--r-- 1 avidullu avidullu 5671 Jun 25 11:46 TRACKNET_WSL_SETUP.md
-rw-r--r-- 1 avidullu avidullu 6077 Jun 25 11:46 UI_ALPHA_EXECUTION_PLAN.md
-rw-r--r-- 1 avidullu avidullu 8922 Jun 25 11:46 UI_PROPOSAL.md
-rw-r--r-- 1 avidullu avidullu 11070 Jun 25 11:46 VIDEO_LOCALITY_MODEL.md
drwxr-xr-x 9 avidullu avidullu 4096 Jun 25 11:46 archives
drwxr-xr-x 2 avidullu avidullu 4096 Jun 25 11:46 data_pipeline
drwxr-xr-x 2 avidullu avidullu 4096 Jun 25 11:46 mockups
---DIR CHECK---
/home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer

```

4. assistant (2026-06-29T10:50:12.858Z)

All three target files exist. Let me read them.

5. user (2026-06-29T10:50:13.926Z)

```

1      # Documentation Status & Archival Register
2
3      > **Status:** ? **LIVING** ? the single board for doc **lifecycle + archival due-
diligence**. **Last full review:** 2026-06-21.
4      > **Update cadence:** whenever a doc changes lifecycle state (created / shipped a
milestone / delivered / archived), and re-review at each clean-slate.
5      > **Why this exists:** so we can (a) **review completeness** at a glance, (b) **update
it as work finishes** to prove we're done, and (c) **never archive a foundational doc without
honestly updating it first** (owner directive, 2026-06-21).
6
7      ---
8
9      ## 0. How to use this
10     - **Reviewing now?** Read §4 (the open findings/actions) ? that's what needs attention.
§3 is the full register.
11     - **Marking something complete?** Update the doc's own **§7 progress tracker** (the
source of truth for *its* completeness), then flip its row here, then ? if terminal ? run the §2
archival checklist.
12     - **This register does NOT duplicate completeness detail** ? it indexes lifecycle +
archive-readiness and points to each tracked doc's §7.
13
14     ## 1. Lifecycle legend
15     `DRAFT ? IN PROGRESS ? DONE/DELIVERED ? ARCHIVED`. Plus: **LIVING** (permanent
reference, never "done"), **PROPOSAL/DESIGN** (owner-gated, not started), **REPORT** (a terminal
record of a run/launch), **CONVENTION** (a standing rule/guide).
16
17     ## 2. Archival due-diligence checklist ? *the repeatable process (do NOT just `git

```

```

mv`)*
18     Before a foundational/tracked doc moves to `docs/archives/`:
19     1. **VERIFY claims against the code** ? no stale assertions; tag load-bearing facts
    `[verified]`. (A doc that says "nothing built" when code shipped is *not* archive-ready ? fix it
    first.)
20     2. **Flip Status ? DELIVERED / DONE / REJECTED + a completion note** ? what shipped,
    where (PRs, paths, artifacts), and what was **deferred / carried forward**.
21     3. **Update ALL inbound references** ? `docs/README.md` index, `CLAUDE.md`,
    `docs/NEXT_STEPS.md`, and any cross-doc links ? the new path + status.
22     4. **Keep the lineage** ? ADR forward/back-links + "supersedes / superseded-by" so the
    evolution stays traceable.
23     5. **`git mv` the whole dir/file ? `docs/archives/past_projects/`** (terminal-state
    ONLY; live docs stay at `docs/` top level). Relative links inside become historical (note this
    in the doc's header).
24     6. **Record the move** ? flip the row here to $3e + a line in memory `current-state`.
25
26     > **Worked example (the template):** **DECOUPLED_COMPUTE ? #270** ? verified M0?M2+M3.5
    delivered on GCP, flipped ??? DELIVERED with a completion note (incl. carried-forward
    WASB-C3/DPA/md5`), reindexed `docs/README` + `past_projects/README`, kept the ADR-010?014
    lineage, then `git mv` to `docs/archives/past_projects/DECOUPLED_COMPUTE/`.
27
28     ## 3. The register
29
30     ### 3a. Tracked projects ? lifecycle docs (archive when DONE; each doc's $7 is the
    completeness source-of-truth)
31     | Doc | Status | Archive-ready? | Next action |
32     |---|---|---|---|
33     | `COMPUTE_DECOUPLED_SERVING/` | **IN PROGRESS** ? M0 approved + **M1?M4 SHIPPED**
    (#279/#280/#282/#283); M5+ default-OFF code batch authorized (2026-06-21) | No | M5 truly-local
    ? M6 Cloud Run ? M8 ledger ? M9-auth ? M11 flywheel ? M12 gdrive-api (owner residual:
    counsel/prices/OAuth/real-run within the D17 cap) |
34     | `RALLY_QUALITY_RESEARCH.md` | Foundation, **tracked & ACTIVE** ($7) | No | Drives the
    R-series; complete when $7.1 single quality number is attributed |
35     | `R2_EVAL_METRIC_DESIGN.md` | IMPLEMENTED (augmenting) | No (gates R4 + dual-eval;
    ablation pending) | Promote-or-reject tolF1 as the gate (ablation) ? then archive with the
    R-cluster |
36     | `R1_MILESTONE_LAUNCH_REPORT.md` | LAUNCHED (#204) ? terminal record | Soon (with the
    R-cluster) | Keep with R-series for narrative; archive when the track wraps |
37     | `RALLY_DETECTION_GAP_CLOSURE.md` | DRAFT (owner-gated) | No | G1?G3 levers; part of
    the R/quality cluster |
38     | `PER_VIDEO_WORKSPACE.md` | DRAFT, IN PROGRESS (P0 = #264) | No | ? supersede-reconcile
    (finding #1) DONE; P3 de-hardcode base ? via #282; remaining P1?P6 (P1 v6 migration coordinates
    with M8's v6 cost-ledger ? version bump authored once) |
39     | `MULTI_SIGNAL_FUSION_PLAN.md` | P1 DONE; P2+ owner-gated | No | P3/P4 (learned fusion,
    audio) ? needs GPU golden features |
40     | `EXPERIMENT_HARNESS.md` | P1 IMPLEMENTED | No | A/B + promotion gate; supports the
    quality track |
41     | `PROMINENT_COURT_DETECTION.md` | DRAFT (owner-gated) | No | C-series; multi-track
    selection design |
42     | `EXECUTION_POLICY.md` | P1 SHIPPED; P3b next | No | Async job policy; reused by Cloud
    Run dispatch |
43     | `GOLDEN_REGRESSION_FIXTURES.md` | IN PROGRESS (owner-gated) | No | Golden eval
    fixtures |
44     | **Data layer** ? `docs/data_pipeline/` (`GOLDEN_SET_IMPLEMENTATION` .
    `GOLDEN_SET_PHASE1_VIDEOS` . `GOLDEN_VIDEOS` . `VIDEO_RECORDING_GUIDELINES` .
    `GOLDEN_DATA_SHARING`) | LIVING (data, **no-ML**) | n/a | Relocated 2026-06-21 to the isolated
    data pillar; tracked via `data_pipeline/README.md`. Roora *vendor* docs archived ?
    `archives/roora-vendor/`. |
45     | `CODE_CLEANUP_AUDIT_2026-06-24.md` | **IN PROGRESS** ? Phase 1: P0/P1/P4/P6a shipped
    (#352/#355/#354/#356). Quick wins (Q4/Q5/Q6/Q8) next. | No | Q4?Q5?Q6?Q8 ? P6b ? P2 (segmenters)
    ? P3 (extra=forbid) ? P5 (app-factory) ? P7 verify |
46     | `DESKTOP_APP_PLAN.md` | DRAFT (owner-gated, nothing built) | No | P0 compute-
    feasibility spike; relates to truly-local profile |
47     | `OWNED_MODEL_IMPLEMENTATION_PLAN.md` | PLAN/roadmap (owner-gated) | No (authoritative
    roadmap) | Gen-0?Gen-2; the owned-model track |

```

```

48 | `PERSONALIZATION_PLAN.md` | DESIGN ? owner-adopted; Phase-0 | No | Per-user models (DB
v5 store landed) |
49 | `UI_ALPHA_EXECUTION_PLAN.md` | ACTIVE | No | khelsutra UI alpha track |
50 | `I18N_PLAN.md` | Design (P0); **in implementation** (adapted) | No | Executed in
`khelsutra/docs/FRIENDLY_AND_MULTILINGUAL.md` (P1?site/, P3?web/); P2 backend stays here, owner-
gated |
51 | `POST_166...RISK_ASSESSMENT.md` | Plan for owner review | No | Milestones/risk doc |
52
53 ### 3b. Foundational / strategy references (LIVING ? archive only if **superseded**, not
when "done")
54 | Doc | Status | Note |
55 |---|---|---|
56 | `PLATFORM_ARCHITECTURE.md` | Strategy/research | The `ComputeBackend`/`StorageBackend`
registries ? realized by COMPUTE_DECOUPLED_SERVING |
57 | `COMMERCIALIZATION.md` | ? LIVING | Strategy; carries WASB-C3 |
58 | `VIDEO_LOCALITY_MODEL.md` | THINKING DOC | L0 fully-local = the truly-local profile |
59 | `HOSTING_PLAN.md` | PROPOSAL | **? partly overtaken by ADR-013 + the shipped site ?
verify (finding #5)** |
60 | `UI_PROPOSAL.md` | PROPOSAL (partly shipped) | UX half **in implementation** ?
`khelsutra/docs/FRIENDLY_AND_MULTILINGUAL.md`; currency banner added (finding #5 ? for this doc)
|
61 | `STORAGE_SHARING_MODEL.md` | DRAFT (co-design pending) | ? |
62 | `ANALYZERS_AND_RECONCILERS.md` | DESIGN (owner-gated) | ? |
63 | `ROADMAP.md` / `CODE_MAP.md` / `HOW_RALLY_DETECTION_WORKS.md` /
`KHELSUTRA_PRODUCT_OVERVIEW.md` | LIVING references | Keep current; not archive targets |
64
65 ### 3c. Reports & terminal records (archive candidates once their track wraps)
66 | Doc | Status | Note |
67 |---|---|---|
68 | `REAL_FOOTAGE_VALIDATION_REPORT.md` | validation complete | Motivates R2; archive with
the R-cluster |
69 | `RALLY_DETECTION_QUALITY_REPORT.md` | report | Quality snapshot; part of R-cluster |
70 | `QUALITY_ITERATIONS.md` | LIVING empirical log | **Keep** ? the paper-backbone
ablation log ([[paper-coauthor-aim]]) |
71 | `CODE_AUDIT_AND_TEST_HARDENING.md` . `HARDENING_LOOP_HANDOFF.md` .
`DEFERRED_HARDENING_PLAN.md` | COMPLETED ? **top-level pointer stubs** (archived copies exist) |
**? confirm intentional stubs (per #165) vs finish-archiving (finding #4)** |
72
73 ### 3d. Conventions / guidelines / templates / runbooks (permanent ? do not archive)
74 `PROJECT_DOC_TEMPLATE.md` . `NEXT_STEPS.md` (? refreshed 2026-06-21) . `BACKLOG.md` .
`EVALUATION.md` . `ABLATION_LAYER.md` . `SETUP_NEW_MACHINE.md` . **`CREATING_REELS.md`** (? end-
user usage guide ? install ? process ? pick ? reel; LIGHT tier; added 2026-06-23) .
**`CONVERT_GCS_VIDEOS_TO_HIGHLIGHTS.md`** (? operator runbook ? `gs://` videos ? reels on L4
Cloud Run; verified e2e 2026-06-23) . `TRACKNET_WSL_SETUP.md` . `README.md` . *(2026-06-21
relocation: the data-layer guides moved ? `data_pipeline/`; the field-ops + Roora docs ?
`archives/{field-pmf,roora-vendor}/`.)*
75
76 ### 3e. Already archived (`docs/archives/`)
77 `past_projects/DECOUPLED_COMPUTE/` (? 2026-06-21, #270) . `past_projects/low-regret-
moves/` . **`field-pmf/`** (field-ops + customer-feedback, 2026-06-21) . **`roora-vendor/`**
(annotation-vendor engagement, 2026-06-21) . `PER_VIDEO_OUTPUT_LAYOUT-SUPERSEDED-2026-06-21.md`
. `CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md` . `DEFERRED_HARDENING_PLAN-
COMPLETED-2026-06-14.md` . `HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md` .
`COMMERCIAL_READINESS_REVIEW.md` . `MONETIZATION_AUDIT.md` . `TEST_RUN_SUMMARY.md` . `research/`
. `checkpoints/` . `quality-iterations/`.
78
79 ### 3f. ADR log ? `docs/archives/decisions/DECISIONS.md`
80 **ADR-001 ? ADR-014** (14, all ratified). The compute/serving lineage: **ADR-005 ? 008 ?
009 ? 010 ? 011 ? 012 ? 013 ? 014** . ADR-014 was **RATIFIED 2026-06-21** (owner M0 sign-off for
COMPUTE_DECOUPLED_SERVING).
81
82 ## 4. Open due-diligence findings / actions (2026-06-21 review)
83
84 > **? Doc-restructuring done (2026-06-21, owner-directed):** field-ops/customer-feedback
? `archives/field-pmf/`, Roora annotation-vendor ? `archives/roora-vendor/`, and a no-ML **data

```

layer** carved into `docs/data\_pipeline/`. Inbound + outbound links rewired; three dir READMEs added (the `data\_pipeline` one sets the "no segmentation/stitching" boundary).

85 1. **RESOLVED (2026-06-21) ?** `PER\_VIDEO\_WORKSPACE.md` supersedes
 `PER\_VIDEO\_OUTPUT\_LAYOUT.md`. **Folded forward the `video\_slug` naming option (workspace §8.5) + the `output/GX010125\_run/` manual-prototype precedent; predecessor archived ?**
 `docs/archives/PER\_VIDEO\_OUTPUT\_LAYOUT-SUPERSEDED-2026-06-21.md`; `CODE\_MAP.md` repointed.

86 2. **RESOLVED (2026-06-21) ?** `NEXT\_STEPS.md` refreshed** with a current 2026-06-21 state + a **cross-track prioritized pending list**; the 2026-06-13 owned-model/quality body is kept (clearly marked) as that track's detail.

87 3. **R-series / quality cluster is ACTIVE** (`R1`, `R2`, `RALLY\_DETECTION\_GAP\_CLOSURE`, `RALLY\_QUALITY\_RESEARCH`, `QUALITY\_ITERATIONS`, `REAL\_FOOTAGE\_VALIDATION\_REPORT`, `RALLY\_DETECTION\_QUALITY\_REPORT`). **Do NOT archive yet** ? archive as a **cluster** once R2 is promoted-or-rejected as the gate + §7.1's single number is attributed. (Owner already confirmed "leave as is", 2026-06-21.)

88 4. **Confirm the COMPLETED top-level stubs** (`CODE\_AUDIT\_AND\_TEST\_HARDENING`, `HARDENING\_LOOP\_HANDOFF`, `DEFERRED\_HARDENING\_PLAN`) are intentional pointer-stubs (per #165) ? or finish-archive them so there's one copy.

89 5. **HOSTING_PLAN.md / UI_PROPOSAL.md** are PROPOSAL but partly overtaken** by ADR-013 (DO dropped) + the shipped khelsutra site. **UI_PROPOSAL.md** banner added (2026-06-21)** ? its UX half is now tracked in `khelsutra/docs/FRIENDLY\_AND\_MULTILINGUAL.md`; the `I18N\_PLAN.md` localization track is likewise in implementation there. **HOSTING_PLAN.md** banner still pending** (ADR-013 already noted in it).

90 6. **RESOLVED (2026-06-21) ?** `COMPUTE\_DECOUPLED\_SERVING` M0 approved + M1?M4 SHIPPED** (#279/#280/#282/#283); all §8 gating Qs locked (D7?D17); M5+ default-OFF code batch authorized (owner residual: counsel ToS/DPA, real GCP prices, OAuth app, real cloud run within the D17 \$100/?10k cap).

91

92 **## 5. Maintenance**

93 Update a row when its doc changes lifecycle; run §2 before any archive; re-review §4 at each clean-slate. This register + the §2 checklist are the operational form of the `CLAUDE.md` archive policy and [[working-agreement-tracked-project-docs]].

94

6. user (2026-06-29T10:50:14.923Z)

1 # Local, Cheap, AI Rally Detection for Racket Sports ? Quality Report (Checkpoint)

2

3 **Internal · honest measured results only · checkpoint 2026-06-17 · supersedes the 2026-06-14 draft**

4 **Project: badminton-highlight-indexer · toward a paper**

5

6 > **An honest empirical study of a shuttle-trajectory rally segmenter, its over-segmentation fix, a**

7 > **task-faithful evaluation metric, a disciplined small-N promotion gate, and a first real-footage**

8 > **validation against human ground truth and a cloud VLM.**

9

10 ---

11

12 **## Abstract**

13

14 We present a fully local, low-cost rally-detection system for racket sports (badminton first) that

15 segments a match video into rally `[start, end]` intervals. The system is built on a **frozen**

16 shuttle-trajectory substrate** (a WASB CNN that emits per-frame shuttle position), from which a

17 velocity signal seeds recall-oriented candidate windows that a tiny pure-numpy logistic-regression

18 scorer grades. Every additional cue enters as a confidence-weighted **feature column** the scorer

19 learns to weight ? never a hand-coded branch ? and ships **default-OFF**, promoted only on a measured

20 held-out lift. We evaluate against an owned golden corpus, now **n=11 labelled matches** (grown from

21 n=6 this cycle, including **5 real owner-hand-labelled ground-truth clips**), treating
the labelling
22 vendor as an interchangeable oracle: the contract is with the golden labels, not their
provenance.
23
24 Since the last checkpoint (2026-06-14) the headline result moved from "one promoted-
then-reverted cue"
25 to a **statistically significant over-segmentation fix**. Bounded windowing plus a
rally-END
26 inactivity trim (the "cap=6 + R4" milestone) lifts our **task-faithful tolerance-match
F1 from 0.091
27 to 0.323** on n=11 (? +0.232, 95% CI [+0.153, +0.304], paired sign-test **10/0,
p=0.002**), drives the
28 **merge-rate from 0.694 to 0.150** (rallies-swallowed-by-a-merge, down on **all 11**
videos), and
29 tightens the mean rally-END error from **52.1 s to 3.8 s**. A first ground-truth
comparison against a
30 cloud VLM (Gemini) shows local reaching **~86% of Gemini on rally detection**
(F1@0.25) with the
31 rally-**START** already at parity** ? the entire residual gap is the rally-END. Negative
and null
32 results are reported in full (person and audio fusion cues remain non-significant at
n=6; a
33 net-crossing gate that wins offline does **not** transfer to the served operating
point). We also
34 introduce a deliberately honest **task-faithful evaluation metric** (asymmetric
tolerance-match with
35 strict 1:1 assignment) after finding that plain temporal-IoU is the wrong ruler for this
task in both
36 directions. The contribution is less any single number than a **method for being honest**
at small
37 scale** ? and the path to a publishable result is now a short one: more labelled
matches.
38
39 ---
40
41 ## Headline results at a glance
42
43 | | |
44 |---|---|
45 | **Over-segmentation milestone** (n=11, R2 tolerance-F1) | **0.091 ? 0.323** . ? +0.232
. CI [+0.153, +0.304] . sign-test **10/0**, p=0.002 |
46 | **Merge-rate** (golden rallies swallowed by a merge) | **0.694 ? 0.150** (down on all
11 videos; p=0.001) |
47 | **Mean rally-END error** | **52.1 s ? 3.8 s** |
48 | **Real-footage ground truth** (n=2) | local **~86 % of Gemini** on detection
(F1@0.25); rally-**START** **at Gemini parity** |
49 | **Gemini cloud reference** | clean **0.93** / multi-court **0.66** (the owned-model
ship target is the 0.66) |
50 | **Golden corpus** | **n=11 matches** (grown from 6), incl. 5 real ground-truth clips;
minors excluded; no biometric labelling |
51 | **Promotion status** | additional cues/tiers (w2 net-crossings, hard-cap, blob-
fallback, person, audio) **default-OFF**; the cap=6 + R4 milestone config flip **LAUNCHED** to the
served default** (PR #204, 2026-06-18) |
52
53 ---
54
55 ## 1. Introduction & Problem
56
57 A rally is the unit of play in racket sports: the span from serve contact to the point
ending.
58 Indexing a long match video into its rallies is the substrate for everything a coach or
player wants ?
59 highlight reels, "show me the longest rally," per-rally statistics. Cloud vision-
language models do

```

60     this well on clean footage, but they are expensive per video, non-deterministic,
network-bound, and ?
61     critically ? **cannot be a training-data source for an owned model under their terms**.
Our constraint
62     is the opposite end of the design space: a system that runs **locally, cheaply,
reproducibly** on a
63     single consumer GPU, with paid LLMs allowed only as an optional inference fallback,
never in the
64     training loop.
65
66     Two failure modes dominate and motivate the method:
67
68     1. **Held-shuttle false positives** ? a player holding or flicking the shuttle between
points produces
69     shuttle/player motion the old velocity heuristic counted as a rally (the owner-
observed precision bug).
70     2. **Over-segmentation / over-merge** ? the segmenter treats "shuttle is moving" as "a
rally is in
71     progress," so dead time and serves **bridge several true rallies into one mega-
window** (the
72     Tier-0 baseline merged **52 % of golden rallies** ; `kushagra_singles` was *2 windows
for 32 rallies*).
73
74     Crucially, the standard temporal-IoU metric is **nearly blind to over-merge** (an
overlap-based match
75     absorbs it), so a method can look fine on IoU while destroying rally structure. Both
failure modes are
76     about the **boundary between play and non-play** ? exactly what this report's windowing
tiers and
77     fusion gates target, and exactly why we had to build an over-segmentation-aware metric
before we could
78     measure the failure we set out to fix.
79
80     ---
81
82     ## 2. System & Method
83
84     ### 2.1 A two-layer mental model
85
86     **Layer 1 ? Detection ("where is the shuttle this frame?").** A frozen WASB CNN emits a
per-frame
87     `(x, y, visible)` shuttle track. The shuttle is the camera-invariant primary signal
because "shuttle in
88     flight" is a near-synonym for "a rally is happening," unlike player motion (which fooled
the old
89     heuristic on broadcast-pan cameras).
90
91     **Layer 2 ? Segmentation ("when is a rally?").** From the per-frame track we compute a
velocity signal
92     that seeds recall-oriented candidate windows; a learned scorer grades each window into
final
93     `[start, end]` rallies. The chain is: `frames ? WASB frozen trajectory ? velocity ?
shuttle-seeded
94     candidate windows ? windowing tiers ? learned scorer over feature columns ? graded
rallies`.
95
96     A key honesty property follows from the data: golden labels are rally **times** (when),
not `(x, y)`
97     coordinates (where). They therefore **cannot retrain the frozen WASB detector** ? they
can only grade
98     and tune the segmentation layer. The discipline is "calibrate on some rallies, measure
on held-out ones."
99
100    ### 2.2 Signals as feature columns, not branches
101

```

```

102     Every producer (detector / analyzer) emits a `(value, confidence)` signal; the scorer
learns to weight
103     those columns and there are no hand-coded `if/else` branches in the decision path. The
scorer is
104     deliberately a tiny pure-numpy **L2 logistic regression** (ADR-004), not a neural net,
so it survives
105     small-N. A cue that does not earn its keep simply stays at weight ? 0 ? it costs nothing
because it is
106     an unused column, not a live branch. Fusion has two realizations: a **decision-level
gate** (label-free,
107     ships first) and a **feature-level path** (the learned route), where cues emit small,
108     scale/translation-invariant per-window features (counts, ratios, zone-membership, two-
sidedness ? never
109     raw coordinates, which would memorize this court and camera).
110
111     ### 2.3 The windowing tiers (this cycle's quality levers)
112
113     All are **default-OFF**, config-gated, and only ever **cut** windows (recall cannot
drop):
114
115     - **W1 ? bounded windowing.** Any window longer than a cap is recursively split at its
largest internal
116     inactivity gap (the See-et-al. sustained-inactivity boundary). Fixes the over-merge.
117     - **R3 ? cap re-scope (12 s ? 6 s).** The rally upper bound; a finer cap measured
strictly better.
118     - **R4 ? rally-END inactivity trim.** After a rally the shuttle keeps drifting slowly
(picked up /
119     walked), over-extending the window END; R4 trims the post-rally tail back to the last
sustained-active
120     frame. START is left untouched (it is already accurate).
121     - **W1.5 / R5 ? hard-cap & blob-fallback.** Last-resort midpoint chops for
*continuously-active* clips
122     with no inactivity gap to split on (a degenerate ~174 s blob). Surgical, do-no-harm,
default-OFF.
123     - **W2 ? net-crossing rally-state gate.** Prunes windows whose shuttle never crossed the
net (the
124     held-shuttle / service-feed discriminator).
125
126     ---
127
128     ## 3. Evaluation Methodology
129
130     This section is the heart of the work: the system's credibility rests on **how** it is
measured.
131
132     ### 3.1 The owned golden corpus and the oracle contract
133
134     Evaluation uses an owned golden corpus, **grown from n=6 to n=11 this cycle** by adding
5 real
135     owner-hand-labelled ground-truth clips (`mahadevpura_1/_2`, `GX010128`, `GX030094`,
`Badminton_BXH_2`)
136     to the original six (`adarsh_avi_singles, kushagra_singles, targetest_doubles, gbaaddy,
testlarge_short,
137     mahadevpura_singles`; 262 rallies). The corpus is classified Proprietary, **minors
excluded entirely**,
138     no biometric/identity labelling. The labelling vendor (Roora) is treated as an
**interchangeable oracle**:
139     the contract is with the golden labels ? trustable ground truth validated by automated
checks plus a
140     ~20 % human spot-check at  $\pm 0.3$  s boundary tolerance ? not with their provenance.
Evaluation rigor is
141     therefore decoupled from the vendor, and the corpus scales by adding or swapping
oracles. **This is the
142     project's moat: a consented dataset plus an honest eval harness, not the weights.**
143

```

```

144     ### 3.2 Leave-one-video-out, with honest statistics
145
146     Learned variants are scored **leave-one-video-out (LOVO)**, grouped by match: train on
the other videos'
147     candidates, predict the held-out video, score against its labels ? never a held-out
*window* from the
148     same video (windows within a video are correlated and flatter the score). Because a bare
LOVO mean at
149     small N is not trustworthy, every promotion attaches: a **bootstrap 95 % CI**, a
**paired ?F1 CI**, and
150     an **exact paired sign test**. The promotion rule is encoded directly: *a lift is real
only when the
151     ?F1 CI clears 0 and the sign test agrees.*
152
153     ### 3.3 A task-faithful metric (R2) ? tIoU was the wrong ruler
154
155     We found plain temporal-IoU-F1@0.5 is the **wrong ruler in both directions**: for a ~6 s
rally it
156     tolerates only ~±1.9 s, and the intrinsically-fuzzy ~1?1.5 s service window nearly
exhausts that
157     (over-penalizing the START); and its overlap-based matching is murky about **over-
production** (W1
158     over-segments ~1.5× yet tIoU hides it via overlap). We therefore introduced **R2**, a
task-faithful
159     metric: an **asymmetric tolerance-match** ( `|?start| ? ?_start = 2.0 s` AND `|?end| ?
?_end = 1.5 s` )
160     under **strict 1:1 Hungarian assignment** (never greedy overlap), reporting `P@? / R@? /
F1@?`, a
161     **start/end boundary-error split**, an **over-segmentation guard** on normalized
merge/split rates, and
162     a **?-sweep** {1.5, 2.0, 2.5, 3.0}. **tIoU/mIoU are demoted to secondary diagnostics.**
R2 is honest by
163     construction: it *lowers* our headline number (tIoU-F1 0.440 ? tolerance-F1 0.323 on the
milestone) by
164     charging for the over-production and loose ends tIoU absorbed.
165
166     ### 3.4 Over-segmentation-aware harness + dual-eval-set discipline
167
168     `segmentation_metrics.merge_split_report` breaks predictions into matched / **merge** (1
pred ? ?2
169     rallies) / split / missed / spurious, surfacing `merge_rate` / `split_rate` ? the
failure tIoU hid.
170     A **nightly regression tripwire** (CPU-only; WASB trajectories are cached, only the
windowing/scorer
171     re-runs) re-scores the golden corpus under two configs ? **DEFAULT** (served production
windowing, all
172     milestone knobs OFF) and **MILESTONE** (cap=6 + R4) ? and trips on any gated regression.
The owner-adopted
173     **dual-eval-set discipline** maintains a growing GOLDEN-finalized eval set + a RAW eval
set so that a
174     late win cannot silently undo an earlier one.
175
176     ### 3.5 Persist once, re-score forever
177
178     Expensive, non-deterministic per-frame detection (WASB trajectory, person tracks, audio
hits) runs once
179     per video on the GPU and is persisted as a small replayable feature substrate. The
cheap, deterministic
180     decision (windowing, gates, scorer) then re-scores any number of **Variants** offline,
with no GPU and
181     zero production risk ? the reproducibility thesis of the whole harness.
182
183     ---
184
185     ## 4. Results

```

```

186
187     ### 4.1 Reference baselines that drive strategy
188
189     Two reference footages anchor strategy: Mahadevpura (clean-ish, single court) and
Kushagra
190     (multi-court, the hard case). The reading is blunt: clean footage is commoditized (a
Gemini serve
191     reaches 0.93), but multi-court drops the cloud wrapper to 0.66 ? and that 0.66 is the
owned model's
192     ship target, not the clean ceiling.
193
194     Table 1 ? Rally-detection F1 at IoU 0.5 vs human golden labels (per-venue
references).
195
196     | Method | Mahadevpura (clean) | Kushagra (multi-court) |
197     |---|---|---|
198     | Heuristic (velocity windowing) | 0.657 | 0.157 |
199     | Gen-0 owned learned head (LOVO, n=6) | 0.744 | 0.561 |
200     | Two-stage WASB + Gemini refine | 0.83 | ? (not measured) |
201     | Gemini wrapper (`gemini-flash-latest`) | 0.93 | 0.66 |
202
203     The owned Gen-0 head already reaches 0.561 multi-court at n=6 ? a corpus-growth-
sized gap to the
204     0.66 target, not an architecture-sized one. (The aggregate heuristic LOVO floor
across n=6 is
205     0.480; the learned head's aggregate LOVO is 0.626, sign-test 5/6, p=0.109 ? not
yet significant,
206     ship=False.)
207
208     ### 4.2 The over-segmentation milestone (this cycle's headline)
209
210     Bounded windowing (W1), a finer cap (R3, 6 s), and the rally-END inactivity trim (R4)
compose into the
211     "cap=6 + R4" milestone, measured on the full n=11 corpus under the task-faithful
R2 metric.
212
213     Table 2 ? Windowing ladder (n=11, R2 tolerance-F1, ?_start=2.0 s / ?_end=1.5 s).
214
215     | Config | tolerance-F1 (mean [95 % CI]) | tIoU-F1 (secondary) | mean \?end\ |
216     |---|---|---|---|
217     | baseline (all-OFF) | 0.091 [0.026, 0.179] | 0.195 | 52.1 s |
218     | + W1 cap=12 | 0.168 [0.100, 0.241] | 0.357 | 6.5 s |
219     | + cap=6 (R3) | 0.276 [0.174, 0.380] | 0.407 | 5.7 s |
220     | + R4 (milestone) | 0.323 [0.210, 0.433] | 0.440 | 3.8 s |
221
222     Table 3 ? Promotion evidence (paired sign-tests).
223
224     | Comparison | ?tolF1 | 95 % CI | sign-test | significant? |
225     |---|---|---|---|---|
226     | baseline ? milestone | 0.232 | [+0.153, +0.304] | 10/0/1, p=0.002 | ? |
227     | cap=12 ? cap=6 (R3) | +0.108 | [+0.070, +0.154] | 10/0/1, p=0.002 | ? |
228     | cap=6 ? +R4 | +0.047 | [+0.022, +0.072] | 8/0/3, p=0.008 | ? |
229     | W1 alone (n=6, tIoU-F1, served) | +0.139 | clears 0 | 6/0, p=0.031 | ? |
230     | W2 net-crossing gate (n=6, tIoU-F1, served) | +0.019 | [+0.011, +0.027] | 6/0, p=0.031
| ? (golden) / not GT-backed |
231
232     The milestone is the first statistically significant rally-quality win of the
project and ? unlike
233     the earlier Gate-A episode ($4.5) ? it is served-measured (the milestone is a
served windowing
234     config), so there is no eval?serve transfer risk.
235
236     ### 4.3 The over-merge is fixed (the failure tIoU hid)
237
238     The over-segmentation-aware metric shows the pathology directly and its repair:

```

239

240 - **merge-rate** 0.694 ? 0.150** on n=11 ? down on **all 11** videos (? ?0.545, sign-test 0/11, p=0.001).

241 (On the original n=6 served view, W1 alone took merge-rate **0.523 ? 0.038**.)

242 - **mean rally-END error** 52.1 s ? 3.8 s** ? the mega-window collapse is gone.

243 - The **net over-segmentation guard** (a weighted merge/split-rate cost, merges weighted higher because

244 a merge hides a rally = recall loss) **passes** (net cost 1.388 ? 0.423), confirming the structural

245 change reduces real over-segmentation rather than trading it for over-splitting.

246

247 **### 4.4 First real-footage validation against ground truth**

248

249 The capstone of the cycle: the windowing work measured on **6 real Kushagra/Yash clips** (fully-local,

250 no-AI WASB, with Gemini as a strong reference) and ? for the first time ? against **owner-hand-labelled**

251 ground truth** on two of them. The over-merge fix is **real at scale** (aggregate over 6 real clips:

252 windows 48 mega-windows ? 250; mean window length 62.9 s ? 10.8 s; merge-groups 27 ? 10; mean

253 matched-IoU 0.23 ? 0.39). The signature pathology ? `GX010129`: *2 windows / 267 s / 27 rallies

254 trapped* ? is gone. W1 then mildly over-corrects into over-segmentation (~1.5x), and we verified this is

255 **clip-content-driven, not venue-driven** (the "GoPro-vs-court" story is unsupported).

256

257 **Table 4 ? First ground-truth at-par (vs human-labelled rallies).**

258

Clip (golden rallies)	metric	Gemini (cloud)	best LOCAL (no-AI)
---	---	---	---
mahadevpura-2 (40)	F1@0.5 / F1@0.25	0.557 / 0.835	0.296 / 0.722
GX010128 (52)	F1@0.5 / F1@0.25	0.813 / 0.857	0.457 / 0.756

263

264 Local is **~86%88 %** of Gemini on rally detection (F1@0.25)** but only **~53%56 %** on tight boundaries

265 (F1@0.5)**. On GX010128 local even **out-recalls Gemini** ? it overlaps all 52 golden rallies; Gemini

266 matched only 39/52 (missed 13).

267

268 **Table 5 ? The gap is the rally-END; the START is already at Gemini parity (matched-pair median, s).**

269

Clip	local \ ?start\	Gemini \ ?start\	local \ ?end\	Gemini \ ?end\
---	---	---	---	---
mahadevpura-2	1.74	1.46	3.12	1.19
GX010128	0.78	0.98	2.12	0.76

274

275 Local **naails the START** (? or better than Gemini); the **entire residual gap** is the rally-END**

276 over-extending. This crisply localizes the next lever: a stronger sustained-inactivity rally-END rule,

277 **not** a generic boundary head and **not** a serve/START classifier (START is solved).
Honest caveat:

278 mahadevpura-2 hit 1.03x of Gemini's rally **count** yet F1@0.5 only 0.150 with W1+W2 ? right count, wrong

279 boundaries. **Count parity** is a deceptive proxy; ground truth is the only honest judge.**

280

281 **### 4.5 The Gate-A eval?serve finding (reported in full)**

282

283 A net-crossing gate (`net\_crossings ? 2`) was promoted offline (?F1 **+0.074**, 6/6, p=0.031, CI

284 [+0.042, +0.104]) and briefly flipped to the served default ? then **reverted**. On the served **coarse**

285 windowing the gate is ****neutral-to-negative** (mean served ?F1 ?0.008, ~12 real rallies dropped)**. The

286 root cause is canonical: the offline harness used a recall-oriented ****proposal****

287 windowing that

288 over-segments (so the gate had abundant junk to cut), whereas the served windowing already merges

289 aggressively (so the gate trims ****real**** rallies instead). The honest, served replacement is ****W2****

289 (§4.2): on the now-bounded W1 windowing the same net-crossing signal is a small but robust ****+0.019**

290 (6/0)** win ? a satisfying reversal that confirms ***the cue's value is windowing-dependent***. A standing

291 litmus (``served_gate_a.py``) pins this behaviour and trips if a future change silently revives or breaks

292 it. ****We do not count the reverted +0.074 as a live win.****

293

294 **### 4.6 Fusion ablation: person and audio (null / uncertain at n=6)**

295

296 GPU person-feature extraction finished all 6 original golden videos; the LOVO drivers measured the lift

297 with honest stats. Neither cue clears the bar (Table 6) ? both stay ****default-OFF****.

298

299 ****Table 6 ? Fusion ablation (n=6, honest LOVO, IoU 0.5).****

300

Variant	F1	?F1 vs prior	95 % CI	sign-test	clears 0?
shuttle-only	0.307	? ? ? ?			
+ person (P3)	0.286	?0.021	[?0.165, +0.162]	2W/4L, p=0.688	No
+ person + audio (P4)	0.366	+0.079	[?0.054, +0.203]	4W/2L, p=0.688	No

307 Person adds noise (without court polygons, the in-court count includes spectators). Audio is the more

308 promising lead (wins 4/6; +27.7 % point estimate) but n=6 cannot confirm it. ****We report these**

309 **deliberately**** ? negative and uncertain results are the honest ablation backbone.

310

311 **### 4.7 The learned-variant operating-point fix**

312

313 An early window-level logistic variant lost to control and zeroed a video ? a fixed-0.5-cutoff artifact,

314 not a real defect. Adding an additive, default-OFF ****threshold-tuning** step (operating point chosen on

315 the train fold)** plus probability calibration (Platt / isotonic) lifted mean F1 ****0.307 ? 0.423****

316 (+0.116), recovered ``testlarge`` from 0.000 to 0.300, and the learned variant now beats CONTROL (0.388).

317 The lesson ? ***calibrate and tune the operating point inside the LOVO loop*** ? is now baked into the harness.

318

319 ---

320

321 **## 5. Detailed comparison with Gemini**

322

323 Gemini (``gemini-flash-latest``) is our ****reference and optional inference fallback ? never a training**

324 source, never the owned target on clean footage**. The comparison is multi-dimensional, and "who wins"

325 depends entirely on the axis.

326

327 ****Table 7 ? Owned local system vs Gemini cloud wrapper.****

328

Dimension	Gemini wrapper (cloud)	Owned local system
Accuracy, clean (Mahadevpura)	**0.93**	F1 (P 1.00 / R 0.87 / mIoU 0.80 / boundary 0.7 s)

329 | Gen-0 0.744; windowing milestone closes most of the over-merge |

332 | Accuracy, multi-court (Kushagra) | ****0.66**** F1 (P 0.69 / R 0.62 / mIoU 0.73 / boundary
 1.2 s) | Gen-0 ****0.561**** ? 0.10 behind, corpus-sized |
 333 | Rally ****detection**** (F1@0.25, ground truth) | reference | ****~86?88 % of Gemini****; out-
 recalls it on GX010128 (52/52 vs 39/52) |
 334 | Rally ****START**** boundary | parity | ****at parity**** (\|?start\| 0.78?1.74 s vs 0.98?1.46
 s) |
 335 | Rally ****END**** boundary | tight (\|?end\| ~0.8?1.2 s) | the ****gap**** (\|?end\| 2.1?3.1
 s) ? the next lever |
 336 | Cost | reference | ****< \$1 for both reference videos**** at flash pricing (serial, ~14
 chunk requests); ****\$0**** for the local path |
 337 | Determinism | ****non-deterministic**** (VLM; outputs vary run-to-run) | ****fully**
deterministic** ? re-scoring yields byte-identical intervals |
 338 | Reliability | ****503 capacity waves throughout**** (one chunk needed ~5 min backoff); a
 cold/prepaid project burst-throttles with a misleading "credits depleted" 429 | local; no
 external dependency. ****Sole-provider flakiness is why the provider-registry fallback is load-**
bearing** |
 339 | Privacy / offline | requires upload + server-side processing + live network | ****100 %**
local**; CPU-only for the decision layer after one GPU pass |
 340 | Training-data licensing | ****barred**** ? paid-LLM outputs can never train the owned
 model (terms/copyright) | owned corpus + vendored labels only; ****clean MIT/Apache/BSD**** weights
 path |
 341
 342 ****The strategic reading.**** Clean footage is ****commoditized**** ? for a few cents a Gemini
 call returns
 343 0.93, so we **serve** clean footage with Gemini and reserve owned weights for where they
 matter. The
 344 multi-court ****0.66**** is the line the owned model must reach to be economically
 justified; the Gen-0 head
 345 is already at 0.561 with only n=6. Two-stage WASB+Gemini refine (0.83 clean) is the
 cost-efficient
 346 serving path for long tapes (it uploads only candidate clips). The honest bottom line:
****Gemini wins on**
 347 absolute boundary accuracy today; the owned system wins on cost, determinism, privacy,
 offline operation,
 348 and the ability to be trained and owned ? and it has already reached START parity and
 ~86 % detection
 349 parity on real ground truth.**
 350
 351 ---
 352
 353 **## 6. Limitations**
 354
 355 - ****Sample size.**** n=11 (5 with real ground truth) ? enough for direction-checks, LOVO
 sign-tests, and
 356 now significant windowing results, but not enough to claim a general model. Corpus
 growth is the
 357 highest-ROI unblock.
 358 - ****Person & audio CIs span 0**** at n=6; neither fusion cue is promotable yet.
 359 - ****Court polygons absent**** ? the mechanical reason +person is a wash (counts
 spectators).
 360 - ****Served path partially live-verified.**** The over-merge windowing is served-measured;
 the full fusion
 361 segmenter on live video is verified offline on cached trajectories. (The milestone
 config flip has since
 362 LAUNCHED to the served default ? PR #204, 2026-06-18 ? so it is no longer owner-
 gated.)
 363 - ****Detector domain gap unmeasured.**** WASB/TrackNet and the person detector were trained
 on other rigs;
 364 their accuracy on our amateur/static-tripod footage is not yet quantified, and every
 downstream cue
 365 inherits that gap.
 366 - ****C3 ? WASB checkpoint provenance**** is a commercial-ship blocker a clean head does not
 launder; an
 367 owned or cleanly-licensed detector is required before commercial release.
 368

```

369     ---
370
371     ## 7. Planned work ? next steps & expected wins
372
373     1. **Grow the corpus via the oracle (the #1 lever).** Priority: multi-court badminton
374     (the target
375         distribution), then ?3 matches per new sport (table tennis first ? WASB transfers at
376         F1 0.909 in the
377         literature). **Expected win:** at n=10 with 9/10 wins the exact sign-test reaches **p
378         ? 0.011** ? and
379         we are already at n=11 with 10/0 on the milestone (p=0.002), so the cue-level results
380         become
381         publishable as the corpus and ablations widen.
382     2. **Promote the milestone (cap=6 + R4) to the served default** ? ~~a 2-line config
383     flip, owner-gated~~ **DONE / LAUNCHED (PR #204, 2026-06-18)**: re-verified at n=13 and flipped
384     to the shipped default (`max_window_duration=6.0`, `window_inactivity_end=1.5` in
385     `backend/config/models.py`); the significant, served-measured, no-regression win of this cycle.
386     3. **Close the rally-END gap** ? the ground-truth-localized lever (|?end| 3.8 s ? toward
387     Gemini's ~1 s):
388     strengthen the sustained-inactivity END rule; R4 already bought ?1.9 s (+0.047, 8/0,
389     p=0.008).
390     4. **Activate court polygons.** The intake manifest already collects `court_corners /
391     net_line /
392     singles_or_doubles`; masking off-court persons is the direct fix for the person-cue
393     spectator noise.
394     5. **Re-measure person/audio on the bounded windowing** (the GPU golden-feature LOVO),
395     and measure audio
396     on shuttle-only **directly** (without the person handicap; the E1 probe found ~2.2×
397     discrimination).
398     6. **ActionFormer/ASFormer (MIT) swap for the Gen-0 logistic head** ? $0, minutes on a
399     2070S. *Caveat from
400     the literature:* parameter-free pooling beats learned 1-D conv at small N
401     (TemporalMaxer), which is
402     exactly why the current scorer is a logistic regression ? so this is gated on corpus
403     growth.
404     7. **Owned VLM (Qwen3-VL, Apache-2.0; InternVL3 MIT fallback)** ? TAL-head-first
405     (boundary accuracy is the
406     product; fine-tuned VLMs miss strict boundaries), commercial-clean across code, data,
407     and weights.
408
409     ---
410
411     ## 8. Areas yet to be explored
412
413     - **Detector domain gap** ? quantify WASB/TrackNet and the person detector on our own
414     footage (false
415     negatives, drift); every cue inherits this unmeasured gap.
416     - **Served-path live end-to-end** ? run the fusion segmenter on a real static-tripod
417     clip and confirm the
418     held-shuttle/warm-up/spectator false-positive drop directly (not only on cached
419     trajectories).
420     - **Two-stage refine on multi-court** ? measured at 0.83 clean, **unmeasured** on the
421     multi-court target
422     distribution where the moat lives.
423     - **Continuously-active footage** ? R5 rescues the degenerate blob (0.000 ? 0.238) but
424     blindly; a
425     rally-state-aware split (net-crossings / two-sided motion) is the principled fix.
426     - **Pose / stance (parked)** ? Apache-2.0 options exist but domain shift on court angles
427     is unproven;
428     box-first remains the baseline.
429     - **Formal active learning** ? *which* video to label next (max local-vs-Gemini
430     disagreement + scorer
431     uncertainty + venue/fps diversity) is designed but not yet a method.
432     - **Multisport clean detectors** ? table-tennis WASB transfer is unverified on our
433     footage; pickleball

```

408 needs a clean MIT/Apache ball detector.
409
410 ---
411
412 ## 9. Related work & path to submission
413
414 ****Related work (cite, don't claim).**** The architecture's skeleton is standard and is
cited, not claimed.
415 Candidate-windows-then-learned-scorer is ****temporal action localization**** (proposal-
then-score). The
416 over-segmentation fixes are ****temporal action segmentation**** refinement / post-
processing. "A learned
417 arbiter over cue columns" is ****late / score-level fusion**** realized as stacked
generalization (? 20 years
418 old). In-domain prior art (****See et al. 2024/25**** rally start/end via sustained
inactivity ? which W1/R4
419 directly port; ****Hsu et al. 2024**** trajectory+swing hit detection; MonoTrack;
ShuttleSet) hard-codes the
420 same cues as `if/else` verdicts. *(Citations are reused from the project's own
adversarially-verified
421 literature review; none are introduced unverified here.)*
422
423 ****What is genuinely novel ? and narrow.**** (1) The ****task-faithful asymmetric tolerance-
match metric****
424 with strict 1:1 assignment and an over-segmentation guard, motivated by tIoU's measured
blindness for
425 this task; (2) the ****report-first reconciler-as-linter**** + the ****disciplined small-N
promotion gate****
426 (default-OFF cues, bootstrap CI + paired sign test, promote-only-when-CI-clears-0,
rollback = config
427 flip). Everything else (the fusion mechanism, the persist-once substrate, the no-
regression harness) is
428 textbook, and we claim only the specific discipline.
429
430 ****Honest publishability verdict.**** This is an ****applied / systems**** contribution, not an
algorithmic one
431 ? there is no new loss, architecture, or dataset family. At the previous checkpoint
(n=6, one
432 promoted-then-reverted cue, two null cues) it was ****not**** conference-ready. This cycle
changes that
433 materially: a ****significant**** over-segmentation result at ****n=11 (p=0.002)****, a ****task-
faithful metric****
434 contribution, and a ****ground-truth-backed START-solved / END-gap diagnosis****. The
minimal additions for a
435 credible submission are concrete and within reach:
436
437 - ****Corpus ? n ? 16****, with multi-court isolated as its own held-out slice (don't
conflate clean and
438 multi-court baselines).
439 - ****Lead with the tolerance-match metric + over-segmentation metrics**** (merge-rate,
F1@k), not bare tIoU.
440 - ****Per-cue ablation with CIs**** on the grown corpus (the `ablation.py` harness already
exists).
441
442 ****Target venues**** (applications/systems-oriented): ****CVIU, ICIP, IEEE Trans. Multimedia,
or a CVPR/
443 ICCV workshop**** (CV-for-sports / fine-grained video). Not the CVPR/ICCV main track (no
algorithmic
444 novelty). The cheapest path to a publishable result remains ****more labelled matches****,
not more
445 engineering.
446
447 ---
448
449 ## 10. Engineering & reproducibility

```

450
451     - Experiment leaderboard ? every compare() appends one reproducible row to
452       eval_baselines/experiment_leaderboard.json, pinned to (spec_hash, label_set_hash,
453       IoU), so any
454       regression traces to a recipe change, not drift.
455     - Nightly regression tripwire ? CPU-only (cached trajectories), guards DEFAULT
456       (served) and MILESTONE
457       configs; exit 1 = regression, 4 = stale baseline.
458     - Standing litmus ? served_gate_a.py pins the honest eval?serve facts.
459     - CI gates ? Ruff, lenient-but-green mypy with a ratchet, syntax+import smoke (no
460       GPU stack), and the
461       full offline mocked suite with a coverage floor. Combined with default-OFF cues + a
462       pinned CONTROL
463       variant**, rollback is a config flip, never a git revert.
464     - Privacy-safe fixtures ? golden-regression fixtures + de-attribution tooling allow
465       video-free
466       shipping of the eval sets.
467     ---
468     ## 11. Conclusion
469
470     The contribution of this work is less any single F1 number and more a method for being
471     honest at small
472     scale** ? and this cycle shows that discipline paying off. On an owned golden corpus
473     grown to n=11
474     (now with real ground truth), bounding the windowing and trimming the rally-END produced
475     the project's
476     first statistically significant rally-quality win (tolerance-F1 0.091 ? 0.323,
477     p=0.002), drove the
478     merge-rate down on every video (0.694 ? 0.150), and ? measured against human ground
479     truth for the
480     first time ? put the rally-START at parity with a cloud VLM**, localizing the entire
481     remaining gap to
482     the rally-END**. We reported the nulls in full (person, audio), banked the eval?serve
483     lesson (the
484     reverted Gate-A ? the honest W2), and introduced the task-faithful metric that made the
485     over-production
486     tIoU hid finally visible. The architecture stays deliberately standard where the
487     literature is mature; we
488     claim novelty only on the task-faithful metric and the small-N promotion discipline. The
489     path to a
490     publishable result is now short and concrete: more labelled matches, multi-court
491     isolated, ablations on
492     the grown corpus** ? the moat was always the consented data plus the honest harness, not
493     the weights.
494

```

7. user (2026-06-29T10:50:14.963Z)

```

1     # Next steps (live) ? refreshed 2026-06-21
2
3     > How to read: this top section is the current cross-track state + prioritized
4     pending list** (refreshed 2026-06-21, replacing the stale 2026-06-13 header). The detailed
5     owned-model & rally-quality track plan (2026-06-13) is preserved below, still valid ? it
6     is now one of several active tracks. Live blow-by-blow: memory current-state.md. Doc
7     lifecycle/hygiene: docs/DOC_STATUS.md.
8
9     ## ? Where we are (2026-06-21)
10
11     DECOUPLED_COMPUTE delivered on GCP + archived** (#270); compute = GCP only
12     (ADR-013, DO dropped). COMPUTE_DECOUPLED_SERVING subproject spun up (#272, ADR-014) ?
13     one pure core + local/cloud profiles; owner M0 design sign-off pending. Per-video
14     workspace P0** shipped (#264); doc-status register + archival due-diligence in place (#273).
15     The khelsutra site + per-rally UI ship in parallel (sibling track). The rally-quality /
16     owned-model** track (below) is the core ML work, gated on growing the golden corpus.

```

7

8 ## ? Prioritized pending ? all tracks

9 **P0 ? highest leverage / unblocks the most**

10 0. **? [owner-priority, 2026-06-21] MAKE IT UI-DRIVEN + ALPHA-SHAREABLE ? "a few recordings lying around ? highlights, from a page, no SSH".** Today the cloud flow is **SSH/CLI-only** (`docs/CLOUD\_GPU\_DRYRUN\_GUIDE.md`); **M12 does NOT fix it.** Needs (a separate khelsutra `web/` SPA + deploy track, NOT an engine seam): **(i)** deploy the engine as a reachable ****service**** (M7 ? Cloud Run service, ? owner GCP spend for a **persistent** endpoint), **(ii)** the ****SPA ingest/progress screen**** (khelsutra `web/` B-P2: enter/upload video ? `/api/process` ? poll `/api/jobs` ? reel; ZERO engine code), **(iii)** flip ****M9 edge-auth ON**** (merged #290, default-OFF) to gate alpha tenants, **(iv)** a ****compute PICKER in the UI**** ("your machine vs our cloud", D13/D15). The engine seams make it **possible**; the SPA makes it **usable**. Full record: `[[ui-driven-cloud-serving-gap]]`.

11 0b. **? [packaging track, IN PROGRESS ? 11 PRs merged 2026-06-21/22] Downloadable batteries-included app ? `docs/PACKAGING\_PLAN.md`.** The LIGHT-tier engine is bundle-ready: `pip install` ? `indexer --app` serves the ****bundled SPA**** single-origin, ****torch-free****, with app-home state / ffmpeg resolver / DNS-rebinding guard / single-instance lock / safe DB upgrades (P0a?e, P1a/b, P2-launcher/P2a/P2d/P2e). ****? Next (fresh focused pushes):**** **(1)** P3 ****GPU validation + screencast**** (L4, ~\$0.50, DELETE-after, \$100 cap ? `[[gcp-vm-always-delete]]`; ? native-WASB 0-rally tuned-config blocker), **(2)** P2b-install (uv/script + truly-local default config), **(3)** P2c first-run wizard (cross-repo khelsutra SPA from `/api/capabilities`), then P3-cloud (Cloud Run) + P4 (ONNX/train/annotate = north star). Memory `[[packaging-app-plan]]` / `[[next-session-packaging-app]]`.

12 1. ? ****`COMPUTE\_DECOUPLED\_SERVING` M0?M9 LARGELY SHIPPED**** (2026-06-21, ADR-014 ratified). M1?M4 (#279/#280/#282/#283), ****M5 #286, M6 #287, M8 #288, M9-auth #290**** all merged (default-OFF); the ****real M7 L4 run**** validated the worker (22 rallies, VM deleted, \$0). All \$8 gating Qs locked (D7?D17). ****Owner residual**** (not blocking Claude's default-OFF code): counsel ToS/DPA (M9-consent live + M11 live), real GCP prices+FX (M8 calibrate), Google OAuth app (M12 real), CF team-domain/AUD + ingress-lock; real cloud verification authorized within a ****\$100/?10k cap**** (D17).

13 2. ****[owner, ongoing] Grow the golden corpus**** ? multi-court badminton + ?3 matches/sport (TT first). The single highest-leverage move; the quality track **and** full-corpus training feed on it.

14 3. ****[owner + Claude · GPU] Full-corpus training (n?5)**** ? a real `weights/<ver>/` (the `0.1.0-l4`` bundle is n=2 plumbing-only).

15

16 ****P1 ? near-term, mostly Claude-doable (default-OFF, green-gated)****

17 4. ****[Claude] `COMPUTE\_DECOUPLED\_SERVING` batch ? REMAINING: M11 + M12**** (M5/M6/M8/M9-auth ? merged this session). ****M11**** = `backend/flywheel.py` capture?store?shadow-eval?goldenize plumbing reusing `workspace.py::for_video` + `backend/eval/experiment.compare` + `backend/eval/promotion.py::promote_if_served_safe`, behind `flywheel.*`=OFF + a faked `consent_attested` (? privacy-sensitive ? give it fresh focus). **M12** = backend/storage/gdrive_api.py` `GDriveApiStorage(StorageBackend)` + a `gdrive-api://` scheme (regex already allows the hyphen; add to _KNOWN` + _backend_class` + the StorageConfig Literal) against an INJECTED fake Drive (mirror `gcs.py` DI); keep the mount-only `gdrive://` untouched. ? **M9-consent cols = a new v7 migration** (M8 took v6). M9-consent ToS/DPA text = owner/counsel.`

18 5. ****[Claude] Per-video workspace P1?P6**** (P1 v6 DB migration ? P2 worker ? P3 per-video routing of `detector/`+`tmp/` *`[the `output/chunks_` de-hardcode base = ? done via #282]*` ? P4 ? P5 Launch-Report flip ? P6 GPU parity). ? P1's v6 migration coordinates with the M8 v6 cost-ledger migration ? author the version bump once.`

19 6. ? ****Ops-Agent codified (#291)**** ? `deploy/gcp/create_gpu_vm.sh`` (the one-command VM helper) always bakes the Ops-Agent startup-script; provision.sh grants the APIs+SA roles. PROVEN active on a fresh VM 2026-06-21. (`[[gcp-ops-agent-enable]]`)

20 7. ****[owner] Set the ship-gate target (#172)**** ? needs corpus growth.

21 8. ****[owner-side GPU] R-series: promote-or-reject `R2` tolF1 as the gate**** (the ablation) ? needs GPU-extracted golden features.

22

23 ****P2 ? later / gated / deferred****

24 9. Rally-quality ****P3/P4**** (person-fusion LOVO, audio cue) ? needs GPU golden features.

25 9b. ****[owner · GPU golden] Promote the R4 density fix (finding B / P2 in `RALLY\_DETECTION\_FIXES\_PLAN.md`).**** PR #393 landed it behind `inactivity_temporal_density`` (**\*\*default-OFF\*\***, zero-regression no-op in prod). To ship: golden-set A/B with the flag ON vs OFF at the **\*\*SERVED\*\*** windowing (the eval `windowingPreset`` carries it); flip the config default to True in a one-line PR only if tolF1 doesn't regress. Fixes a sparse-tracking quirk where one

fast sample after a detection gap held the rally tail open (R4 is default-ON in prod).

26 10. ****M0.4 scorer swap**** ? TemporalMaxer (per the 2026-06-17 external review).

27 11. ****Multisport**** (table-tennis first, then pickleball).

28 12. Serving ****M8/M9**** ? per-video cost ledger, edge auth, DPA/consent (within
`COMPUTE\_DECOUPLED\_SERVING`; owner/legal-gated).

29 13. ****WASB-C3**** weight provenance (gates ***sharing*** a trained model) ? legal.

30 14. ****Rename repo off "badminton"**** (owner, BACKLOG).

31 15. Owner housekeeping: retire the shared ****D0 droplet**** + ****D0 token****; collector
****md5**** export (collector repo).

32 16. ****PMF field validation**** (owner-side, business track).

33

34 ****Doc-hygiene (tracked in `docs/DOC\_STATUS.md` \$4):**** ? #1 per-video reconcile + ? #2
this refresh; ****pending:**** field/PMF/Roora ? archives + a data-layer `docs/` dir (owner boundary
confirm); confirm the COMPLETED top-level stubs; verify `HOSTING\_PLAN`/`UI\_PROPOSAL` currency.

35

36 ---

37

38 **## Owned-model & rally-quality track ? detail (2026-06-13 plan; still valid)**

39

40 ****Updated:**** 2026-06-13 (****rally-detection quality track****: fusion + experiment-harness
design docs merged,

41 owner-gated ? see the dedicated section below, the desktop-agent pick-up). Prior:
2026-06-10 (audit?M0?Gemini-battery session; later same day the ****UI-alpha track opened****:

42 ADR-009 ratified ? KhelSutra Guru / khelsutra.guru, private `khelsutra` repo, local-
first delivery ? see

43 `docs/UI\_ALPHA\_EXECUTION\_PLAN.md`; ****PR-1** async jobs (#16) MERGED 2026-06-11, PR #87**;
****PR-2**

44 upload/download MERGED 2026-06-11, PR #99** ? live-E2E-verified (3-part chunked upload
MD5 byte-identity +

45 compile?browser-download); next: PR-3 identity. Same day: the ****8?9** quality sweep
#90?#98 landed** ? ruff/mypy/

46 coverage-floor CI lanes, SQLite conn-close fix, config unknown-key warning, hybrid-twin
collapse into

47 `trajectory\_hybrid.py`, WSL tilde+abspath live-path fixes; suite 388 green; LOVO 0.626
re-verified exact).

48 ****Authoritative roadmap:****

49 `docs/OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md` (+ ADR-008 for topology). ****Latest blow-by-
blow:**** memory

50 `current-state.md` / `session-handoff.md`. This file = the prioritized "what to do
next," refreshed each session.

51

52 **## ? Next-session / GPU-box pickup (2026-06-15)**

53 ****M0** in-container quick wins are COMPLETE:****** ship-gate provisional blocker (****#170****),
train/eval split guard (****#171****), the ****doc-consistency sweep**** (****#174**** ? corrected 16
drifted docs vs the code), and the ****P3 person-feature fusion litmus machinery** is already
present + suite-green** (`backend/eval/fusion\_compare.py` + `ablation.py`: person-driven ?F1 /
bootstrap CI / paired sign test; the real-golden `skipif` litmus waits only on GPU-extracted
data). ? CI is red repo-wide due to a ****GitHub Actions minutes/billing outage**** (account-level,
not code) ? work is local-gated + `--admin`-merged until Actions is restored (Settings ? Billing
? Actions minutes / spending limit).

54

55 ****Immediate next step ? the real P3 LOVO measurement**** (this box has a GPU +
`torch+cu124` + `cv2`; videos and trajectory CSVs are present, but the GPU-extracted golden
features + `.rallies.csv` labels are not yet here):

56 1. ****[owner / GPU box]**** `python -m backend.eval.fusion_golden --manifest
output/human_lovo_manifest.json --out-dir "%USERPROFILE%\OneDrive\rally-annotator\golden-
data\features\2026-06-15-n6" ? produces `\*\_golden\_features.json` for the 6 golden videos. (GPU
+ WSL + videos + trajectory CSVs + `.rallies.csv` labels are owner-side only.)

57 2. Artifacts land in the ****shared cloud folder**** `C:\Users\avidu\OneDrive\rally-
annotator\golden-data\` (OneDrive auto-syncs both ways). Convention + workflow:
****[`GOLDEN\_DATA\_SHARING.md`](data_pipeline/GOLDEN_DATA_SHARING.md)****; tracked in ****#175****.

58 3. ****[CPU dev/agent session]**** `python -m backend.eval.fusion_compare --features-dir
<shared>\features\2026-06-15-n6 --record` + `python -m backend.eval.ablation --features-dir
<shared>\... --granularity group` ? real person-feature ?F1; copy the JSONs into `output/` to
also pass the `skipif` litmus tests (`tests/test\_ablation.py::test\_litmus\_reproduces\_gate\_a`).

```

59
60     **Also queued (owner-gated / pending input):**
61     - Ship-gate target calibration ? **172** (owner decision; needs corpus growth).
62     - Tests reorganization ? blocked on the owner's grouping metric (provide it ? an agent
will do it).
63     - Restore CI ? the Actions outage is account-level (billing/minutes), not a repo/config
bug.
64
65     **Leaving (bulky / M2 ? no start without owner go):** ActionFormer (M0.4), event-index
DB migration (M0.1), Gemini-silver purge (M0.3), cost-to-Gen-2 benchmark + ByteTrack
deprecation, multisport, repo rename, PMF execution.
66
67     ## Where we are (one paragraph)
68     ADR-008 ratified ("repo split driven by productionization, not isolation"): training
lives in-repo (`training/`)
69     behind a CI-enforced import wall; the featurizer is single-sourced
(`backend/features/rally_seq.py`, train==serve);
70     the **Gen-0 harness ships** (`python -m training.gen0.harness --manifest
output/human_lovo_manifest.json --floor
71     eval_baselines/heuristic_lovo_n6.json --version <semver>`) and produced artifact v0.1.0
(LOVO **0.626**, floor passed,
72     sign test 5/6 wins p=0.109 ? honestly not significant, ship=False). The **n=6 owned
corpus is in the collector**
73     (262 rallies, Proprietary, commercial export = owned data only after the ShuttleSet
reclassification). The **Gemini
74     battery is done** (see table) ? the moat is quantified.
75
76     ## The quality table that now drives strategy (IoU 0.5 vs human golden labels)
77     | Path | Mahadevpura (clean-ish) | Kushagra (multi-court) |
78     |---|---|---|
79     | Heuristic (velocity windowing) | 0.657 | 0.157 |
80     | **Gen-0 owned head** (LOVO, n=6) | 0.744 | 0.561 |
81     | Two-stage WASB+Gemini refine | 0.83 | ? |
82     | **Gemini wrapper** (`gemini-flash-latest`) | **0.93** | **0.66** |
83
84     **Reading:** clean footage is commoditized (serve it with Gemini for cents). **Multi-
court drops the commodity
85     wrapper to 0.66 ? that is the owned model's ship target** (its FPs are background-game
pickups + dead-time merges,
86     the exact contrast-metric confound). Gen-0 is 0.10 behind with only n=6 ? a corpus-
growth-sized gap, not an
87     architecture-sized one. Two-stage refine = the cost-efficient serving path for LONG
tapes (uploads candidate clips
88     only); wrapper 503-flakiness (observed all session) makes the provider-fallback registry
load-bearing.
89     Baselines tracked in `eval_baselines/` (heuristic_lovo_n6.json,
gemini_wrapper_2026-06-10.json).
90
91     ## Rally-detection quality track (design merged 2026-06-13, owner-gated; implementation
in progress)
92     **This is the track a desktop/local agent is slated to pick up.** Design docs landed (PR
#112 + session). All implementation is **owner-gated** (per explicit in the plan); each step =
ONE PR off `master`.
93
94     **Hardening loop (T1-T5 positive tests + audit + finale) COMPLETE (2026-06-14, PRs
#156/#157/#160/#161/#163 + #165 + final records).** See archived
`docs/archives/CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md` (Living Log with every
pre/post/review) and `docs/archives/HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md` (full rules
+ STATE). Top-level now pointer stubs per #165 review. All 5 items + verification + archive
done; clean slate.
95
96     - **Docs:** `docs/MULTI_SIGNAL_FUSION_PLAN.md` (shuttle + person + audio, one fusion
layer) .
97     `docs/EXPERIMENT_HARNESS.md` (A/B + shadow + promotion gate, zero-regression) .
98     `docs/ROORA_LABELING_GUIDELINES.md` (v8) . `docs/HOW_RALLY_DETECTION_WORKS.md`

```

(2-layer mental model).

```
99 - **Key finding (no re-investigation needed):** the owner's "held-shuttle counted as a
rally" bug is **NOT a missing capability** ? `backend/pipeline/detectors/rally_gate.py` already
implements the net-crossing **"Gate A" (apply_gate(..., min_crossings=2))` that kills
service-feed/held-shuttle windows, but it is **only called from eval drivers** (and there was no
`min_crossings` knob in `IndexingConfig`). **Wiring Gate A into the live segmenter (now via
fusion_hybrid) was the single cheapest, highest-confidence quality win.** (P2 FusionSegmenter
specifics: registry `fusion_hybrid`, **default-OFF**, seeds from substrate, persists
`net_crossings` + optional person features, optional served Gate A/B via config knobs;
adversarially reviewed; suite green. See `tests/test_served_gate_a_regression.py` for the honest
finding that on production windowing Gate A is neutral/negative in some cases, hence kept opt-in
with knob for safety, and the leaderboard revert note.)
100 - **Current status (2026-06-14):** P1 (person cue extraction to reusable
`PersonDetector`) + P2 (FusionSegmenter + Gate A signal + served handoff gating) + experiment
harness P1 DONE (adversarial review + NO-REGRESSION; suite green). **P3 next (this PR + follow-
ups):** learned fusion (person features ? harness `compare` LOVO lift on the 6 golden videos;
eager-person-compute optimisation); then P4 audio via hit_detector.
101 - **Pick-up order (each = ONE PR off `master`, owner approval first, flag-gated default-
OFF, promote only on measured LOVO):**
102 1. (P2 first-step, largely done via fusion_hybrid) Wire Gate A into production served
path + config knob (min_crossings); apply in hybrids at runtime. (Now available opt-in; served
neutral on production windowing per litmus ? kept opt-in with knob for safety.)
103 2. P3 first step (this work item): harness `compare` LOVO litmus for person-on variant
(players_in_court + two_sided + colocation etc. from `fusion_features`) on the 6 golden; add the
eager compute optimisation note/landing. Pure + harness (testable here).
104 3. P3/P4 continuation: audio cue integration + full learned promotion if lift proven.
105 - **Discipline (from EXPERIMENT_HARNESS.md):** every new cue **flag-gated, default
OFF**; promote to default **only on measured LOVO lift** through `run_regression.py` (exit
0/1/3); pinned `CONTROL` ? rollback = config flip. Harness re-uses ~80% existing code.
106 - ? Gate A, person cue logic, harness, and pure fusion paths are testable here; the
heavier real-video / golden LOVO / ablation confirmation waits on the GPU-extracted golden
features + `.rallies.csv` labels, which are not yet present in this checkout.
107 - **External due-diligence review inputs (2026-06-17) ? PICK UP AFTER current ideas are
exhausted; full details + citations in [`RALLY_QUALITY_RESEARCH.md`](RALLY_QUALITY_RESEARCH.md)
$5.6.** An owner-commissioned 31-reference review of the quality report *validated* the
methodology and surfaced these **future** levers (research inputs, not active work, not new
claims; verify citations before any submission):
108 1. **"First-mile" paper framing (highest-value):** stroke-classifiers (BST/DSTA) + the
big datasets (ShuttleSet ~36k strokes, Fine-Badminton ~27k actions) all **assume a pre-trimmed
rally clip already exists** ? we solve the ignored prerequisite. Pitch the paper as **"the
missing untrimmed-video preprocessing pipeline that lets BST/MonoTrack run on raw footage."**
109 2. **Ground R2 in Action-Spotting / Precise-Event-Spotting**
(SoccerNet/TennisSet/T-DEED) ? turns R2 into a defensible publishable contribution.
110 3. **Add domain baselines for any submission:** MonoTrack + BST on ShuttleSet / Fine-
Badminton (we have only heuristic + Gemini today).
111 4. **Prefer TemporalMaxer over ActionFormer** for the M0.4 scorer swap (parameter-free
max-pool, ~2.8x fewer GMACs, small-N robust) ? see "M0.4 next increment" below.
112 5. **Court polygons ? pose ? the rally-END lever:** masking on-court players can
resolve *occluded end-of-rally states* (the #1 remaining gap vs Gemini), not just spectator
noise.
113 6. **Name the eval?serve lesson "pipeline distribution skew" ? a citable paper
"lesson."
114
115 **Discipline reminder:** owner approval before starting any owner-gated item; one PR per
step; babysit the PR (poll/fix/reply) until merge observed, *then* advance.
116
117 ## Prioritized next steps
118 1. **[owner, in progress] GROW THE GOLDEN CORPUS** ? the single highest-leverage move;
every lever below feeds on it.
119 Priority order for new videos: *(a) multi-court badminton* (the target distribution
AND where the ship target
120 lives), *(b) ?3 matches per new sport ? table tennis first* (WASB transfers at F1
0.909; pickleball labels are
121 corpus-gold but not trainable until a clean MIT/Apache ball detector exists), (c)
capture variety per
```

```

122     `VIDEO_RECORDING_GUIDELINES.md`; **adult sessions only for the training pool**
(consent guardrail). Per video:
123     label ? `curate import-local --normalize --license Proprietary --fps <true fps>` ?
re-run the Gen-0 harness.
124     The paired sign test needs n: at n=10 with 9 wins, p?0.011 ? significance is
reachable this month.
125     2. **[owner decision] Set the ship-gate target** ? now informed: floor = heuristic 0.480
(necessary), **multi-court
126     reference = wrapper 0.66** (the number that makes the owned model worth serving),
clean-footage ceiling = 0.93
127     (not the target ? Gemini serves that tier). Suggested shape: "LOVO F1@IoU0.5 ? 0.70
on multi-court folds with
128     paired-sign p<0.05 vs heuristic" ? owner to ratify/adjust.
129     3. **M0.4 next increment ($0/local):** swap the Gen-0 LogReg for a lightweight temporal
head (MIT, minutes on the
130     2070S ? NOT the heavyweight Gen-2 stack) inside the existing harness; per-video
threshold calibration (the
131     LOVO-vs-oracle gap is visible per fold in the harness output). **Prefer
TemporalMaxer** (parameter-free
132     max-pool, ~2.8x fewer GMACs, small-N robust) over ActionFormer/ASFormer per the
2026-06-17 external review
133     (`RALLY_QUALITY_RESEARCH.md` $5.6) ? better fit for the cheap/local/fast mandate.
134     4. **Multisport thread:** ingest Roora pickleball/TT CSVs (`import-local --normalize`);
merged-prototype LOVO across
135     badminton+TT once TT has ?3 matches; identify a clean pickleball ball detector. ? TT-
on-WASB serving stays gated
136     by C3 (provenance multiplies per sport).
137     5. **PMF validation (cheaper than any engineering month):** C13 academy interviews (+
the two added questions:
138     "multi-court footage nobody watches?" / "pay per-match to index it?"); camera pilot
at the warmest 2-3 academies
139     (consent at capture; demo served via the Gemini path or human-polish ? the reel
pattern exists:
140     `output/MahadevpuraSingles_highlights.mp4`). **Field kit COMPLETE (2026-06-12, owner
ratified the
141     physical-visit approach; reframed product-feedback-first per owner spec):**
142     `docs/PRODUCT_FIELD_ASSOCIATE_FLYER.md` (the recruiting flyer ? owner fills ?/contact
placeholders and recruits) +
143     `docs/FIELD_VISIT_PLAYBOOK.md` (the hired associate's visit playbook: coach + student
script, demo-after-Q6
144     rule, ground rules, Phase-2 record-and-show-back stretch goal) +
`docs/FIELD_VISIT_ANSWER_SHEET.md` (per-visit
145     capture form incl. demo-reaction + student blocks) + `docs/PMF_FIELD_SIGNALS.md`
(G/A/R anchors + **pre-registered
146     decision rules** ? PROCEED / PIVOT A-C / FALSIFIED thresholds fixed before the first
visit).
147     6. **Housekeeping:** ~~rotate the Gemini API key~~ **DONE 2026-06-10** (owner-verified
working; confirm old-key
148     deletion in AI Studio); **rename repo off "badminton"** (owner action,
149     in BACKLOG); collector=SoR confirmation; off-peak niceties: none pending ? the
battery completed.
150
151     ## Pointers / artifacts
152     - Roadmap: `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` · Topology: ADR-008
(`docs/archives/decisions/DECISIONS.md`) ·
153     Study: `docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md` · Quality history:
154     `docs/archives/quality-iterations/2026-06-multivideo-lovo/`.
155     - Harness: `training/gen0/harness.py` (one command ? train ? LOVO ? gate ? artifact
bundle; weights gitignored,
156     manifest reviewable). Featurizer: `backend/features/rally_seq.py` (FEATURE_SCHEMA v1).
157     - Corpus: collector DB (system-of-record, 6 matches/262 rallies) · eval manifest
`output/human_lovo_manifest.json`
158     (gitignored) · durable trajectories + Roora files: `C:\Users\avidu\Projects\Annotation
Setup\Collect\Trajectories\` ·
159     labels: `...\Collect\Golden Labelled`.

```

160 - Gemini ops learnings: serial uploads (`indexing.ai\_max\_workers=1`) + chunk re-encode
161 (`indexing.ai\_chunk\_scale\_width=1280`) + 6-attempt exponential generate retry ? a
cold/prepaid project burst-throttles
162 with a MISLEADING "credits depleted" 429; ramp gradually.
163 - ? No owned-model implementation beyond greenlit M0.x until owner greenlights the next
milestone. Committed next
164 PLATFORM slice = async job model (single-tenant).
165

8. assistant (2026-06-29T10:50:17.822Z)

I have all three files. Let me compile the requested data.

9. user (2026-06-29T10:51:07.667Z)

<tool_use_error>InputValidationError: StructuredOutput was called with input that could not be
parsed as JSON.
You sent (first 200 of 9473 bytes): {"doc_status_protocol": "DOC_STATUS.md archival due-
diligence protocol (the honest update->archive checklist, §2). Run BEFORE any
foundational/tracked doc moves to docs/archives/ (do NOT just `git mv`
Common causes: unescaped backslashes in file paths (use / or \\), unescaped control characters,
or truncated output. Retry with valid JSON.</tool_use_error>

10. assistant (2026-06-29T10:51:10.666Z)

The input must be passed as structured fields, not a wrapped raw JSON string. Let me call
StructuredOutput with the proper field arguments.

11. user (2026-06-29T10:51:50.382Z)

Structured output provided successfully